Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics

Practical – 1

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Class: TYIT	Batch: 1
Date of Assignment: 24/07/26	Date/Time of Submission: 24/07/26

a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.

Code & Output:

Step 1: imports all necessary files.
Step 2: load the data set and display the first & last 5 rows.
Step 3: display the shape of dataset and column name & information.
Step 4: Display data type and check the missing value & duplicate records.
Step 5: Display dscript static, count uniq value & display unique value of particular columns

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load dataset
df = pd.read_csv("Retail_sales.csv")
```

```
#Display First 5 Records
df.head()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount
0	Spearsland	52372247	2022-01-01	9	2741.69	
1	Spearsland	52372247	2022-01-02	7	2665.53	
2	Spearsland	52372247	2022-01-03	1	380.79	
3	Spearsland	52372247	2022-01-04	4	1523.16	
4	Spearsland	52372247	2022-01-05	2	761.58	

```
#Display Last 5 Records
df.tail()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Disco
29995	Spearsland	50239115	2022-01-25	5	2501.15	
29996	Spearsland	50239115	2022-01-26	3	1500.69	
29997	Spearsland	50239115	2022-01-27	6	3001.38	
29998	Spearsland	50239115	2022-01-28	5	2501.15	
29999	Spearsland	50239115	2022-01-29	3	1425.66	

```
#Display Shape of Dataset
df.shape
```

```
(30000, 11)
```

```
#Display Column Names
df.columns
```

```
Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',
       'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',
       'Product Category', 'Day of the Week', 'Holiday Effect'],
      dtype='object')
```

```
#Display Dataset Information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 11 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Store ID               30000 non-null  object
 1   Product ID             30000 non-null  int64
 2   Date                   30000 non-null  object
 3   Units Sold             30000 non-null  int64
 4   Sales Revenue (USD)    30000 non-null  float64
 5   Discount Percentage    30000 non-null  int64
 6   Marketing Spend (USD)  30000 non-null  int64
 7   Store Location         30000 non-null  object
 8   Product Category       30000 non-null  object
 9   Day of the Week        30000 non-null  object
 10  Holiday Effect         30000 non-null  bool
dtypes: bool(1), float64(1), int64(4), object(5)
memory usage: 2.3+ MB
```

```
#Display Data Types
df.dtypes
```

```
Store ID                  object
Product ID                 int64
Date                      object
Units Sold                 int64
Sales Revenue (USD)      float64
Discount Percentage        int64
Marketing Spend (USD)      int64
Store Location            object
Product Category          object
Day of the Week           object
Holiday Effect              bool
dtype: object
```

```
#Display Descriptive Statistics
df.describe()
```

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)
count	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000
mean	4.461294e+07	6.161967	2749.509593	2.973833	49.944033
std	2.779759e+07	3.323929	2568.639288	5.974530	64.401655
min	3.636541e+06	0.000000	0.000000	0.000000	0.000000
25%	2.228600e+07	4.000000	882.592500	0.000000	0.000000
50%	4.002449e+07	6.000000	1902.420000	0.000000	1.000000
75%	6.559352e+07	8.000000	3863.920000	0.000000	100.000000
max	9.628253e+07	56.000000	27165.880000	20.000000	199.000000

```python
#Check Missing Values
df.isnull().sum()
```

```
Store ID                0
Product ID              0
Date                    0
Units Sold              0
Sales Revenue (USD)     0
Discount Percentage     0
Marketing Spend (USD)   0
Store Location          0
Product Category        0
Day of the Week         0
Holiday Effect          0
dtype: int64
```

```python
#Count Unique Values
df.nunique()
```

```
Store ID                   1
Product ID                42
Date                     731
Units Sold                32
Sales Revenue (USD)     2545
Discount Percentage        5
Marketing Spend (USD)    200
Store Location           243
Product Category           4
Day of the Week            7
Holiday Effect             2
dtype: int64
```

```python
#Check Duplicate Records
df.duplicated().sum()
```

```
np.int64(0)
```

```python
#Display Unique Values of a Particular Column
df["Product Category"].unique()
```

```
array(['Furniture', 'Electronics', 'Groceries', 'Clothing'], dtype=object)
```

Step 6: count product category and count store location, holliday effect.

Step 7: calculate sale revenue , maximum & minimum sale revenue , correlation matrix.

```python
#Count Product Categories
df["Product Category"].value_counts()
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```python
#Count Store Locations
df["Store Location"].value_counts()
```

```
Store Location
Korea                       237
Congo                       234
Turks and Caicos Islands    156
Guam                        153
Tanzania                    151
                           ...
Trinidad and Tobago          98
Haiti                        96
Dominica                     95
Ireland                      92
Andorra                      80
Name: count, Length: 243, dtype: int64
```

```python
#Count Holiday Effect
df["Holiday Effect"].value_counts()
```

```
Holiday Effect
False    29836
True       164
Name: count, dtype: int64
```

```python
#Calculate Average Sales Revenue
df["Sales Revenue (USD)"].mean()
```

```
np.float64(2749.509592666667)
```

```python
#Find Maximum Sales Revenue
df["Sales Revenue (USD)"].max()
```

```
np.float64(27165.88)
```

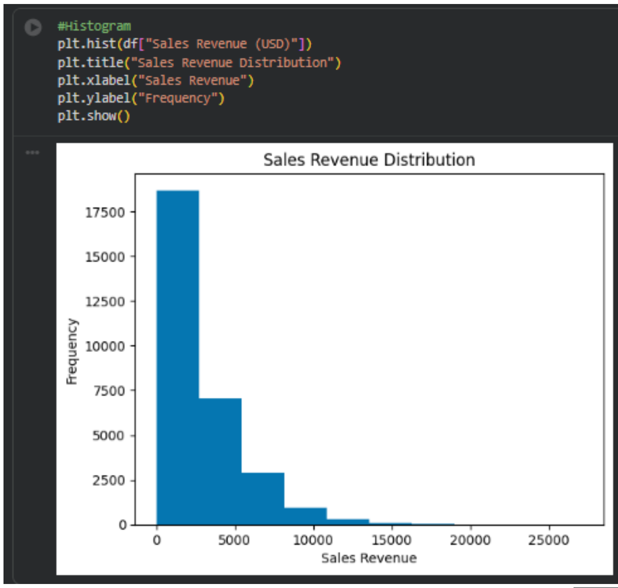
```python
#Find Minimum Sales Revenue
df["Sales Revenue (USD)"].min()
```

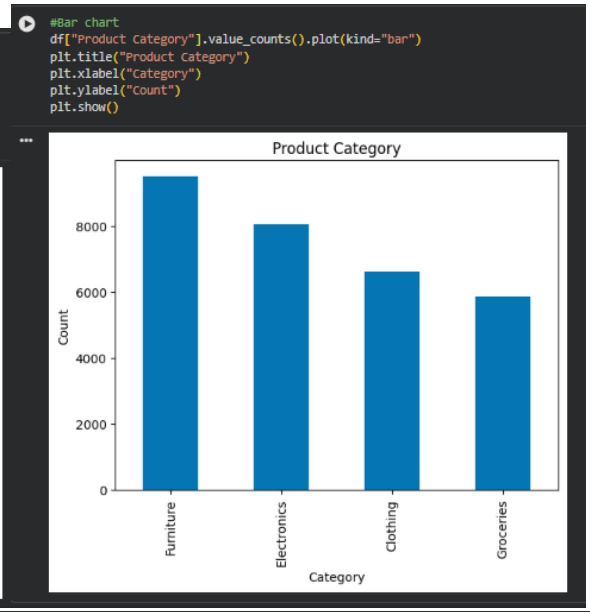
```
np.float64(0.0)
```

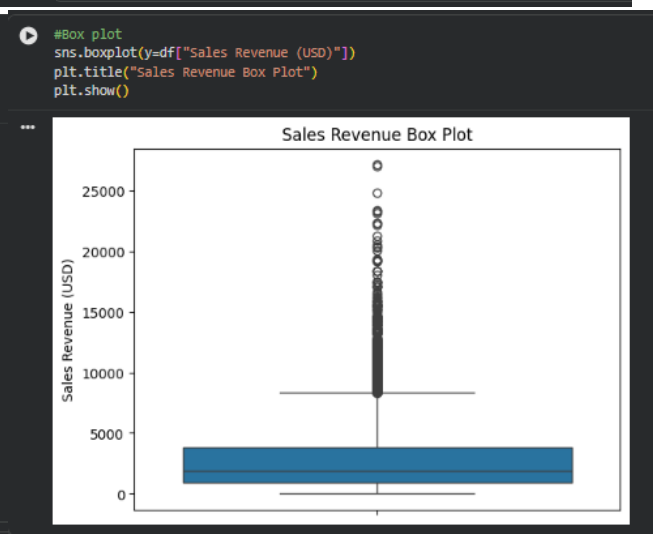
```python
#Correlation Matrix
df.corr(numeric_only=True)
```

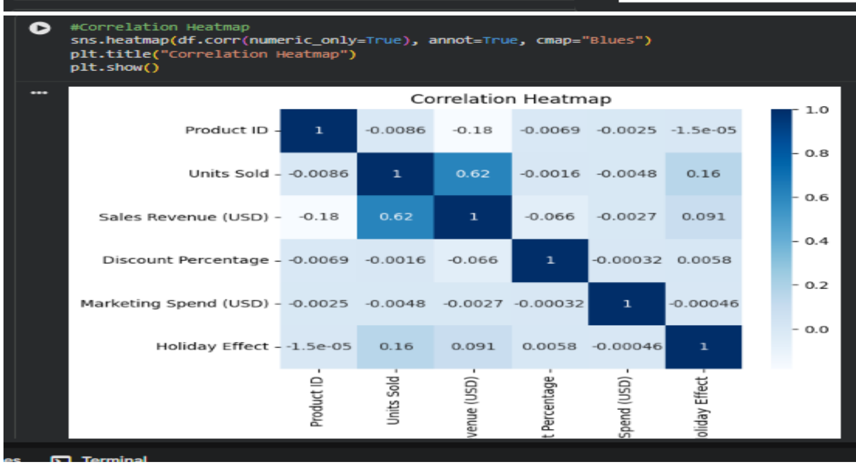
	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Holiday Effect
Product ID	1.000000	-0.008570	-0.181934	-0.006891	-0.002496	-0.000015
Units Sold	-0.008570	1.000000	0.615767	-0.001620	-0.004750	0.159324
Sales Revenue (USD)	-0.181934	0.615767	1.000000	-0.065791	-0.002732	0.090681
Discount Percentage	-0.006891	-0.001620	-0.065791	1.000000	-0.000322	0.005848
Marketing Spend (USD)	-0.002496	-0.004750	-0.002732	-0.000322	1.000000	-0.000455
Holiday Effect	-0.000015	0.159324	0.090681	0.005848	-0.000455	1.000000

Step 8: Make a histogram , Bar, pie , Box and correlation heatmap.









b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

Code & Output:

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Step 1: Import Required Libraries

Step 2: load the dataset

Step 3: Analysis 1 - Product category analysis

```python
#Import Required Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

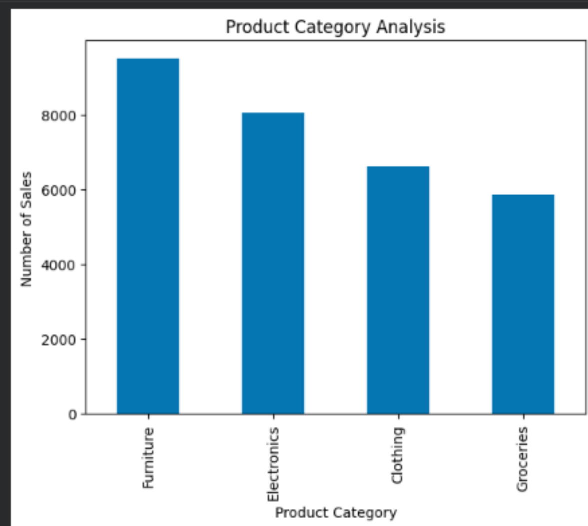
```python
#Load dataset
df = pd.read_csv("Retail_sales.csv")
```

Analysis 1 - Product category analysis

```python
print(df["Product Category"].value_counts())
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```python
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category Analysis")
plt.xlabel("Product Category")
plt.ylabel("Number of Sales")
plt.show()
```



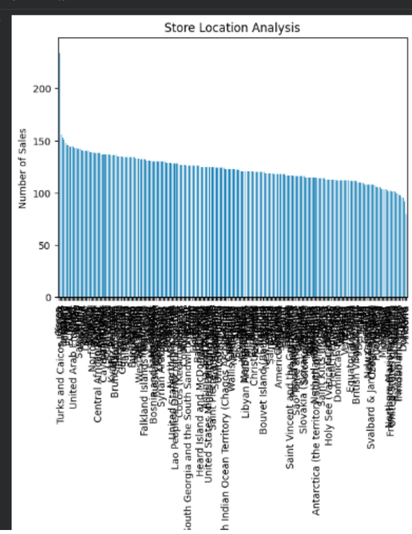
Step 4: Analysis 2 - Store Location Analysis

Analysis 2 - Store Location Analysis

```python
print(df["Store Location"].value_counts())
```

```
Store Location
Korea                         237
Congo                         234
Turks and Caicos Islands      156
Guam                          153
Tanzania                      151
                             ...
Trinidad and Tobago            98
Haiti                          96
Dominica                       95
Ireland                        92
Andorra                        80
Name: count, Length: 243, dtype: int64
```

```python
df["Store Location"].value_counts().plot(kind="bar")
plt.title("Store Location Analysis")
plt.xlabel("Store Location")
plt.ylabel("Number of Sales")
plt.show()
```



Step 5: Analysis 3 - Day of the Week Analysis & Analysis 4 - Holiday Effect Analysis

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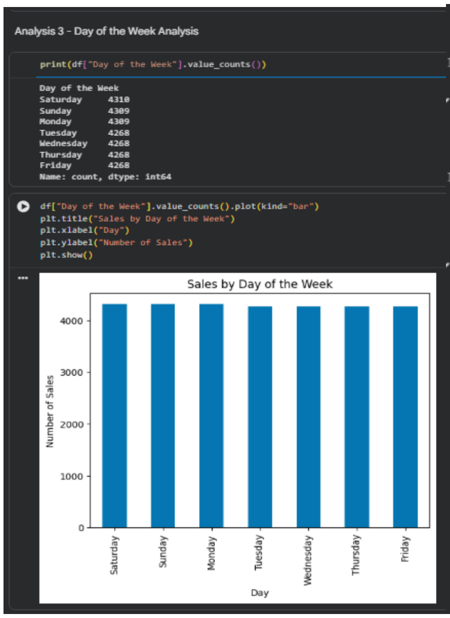
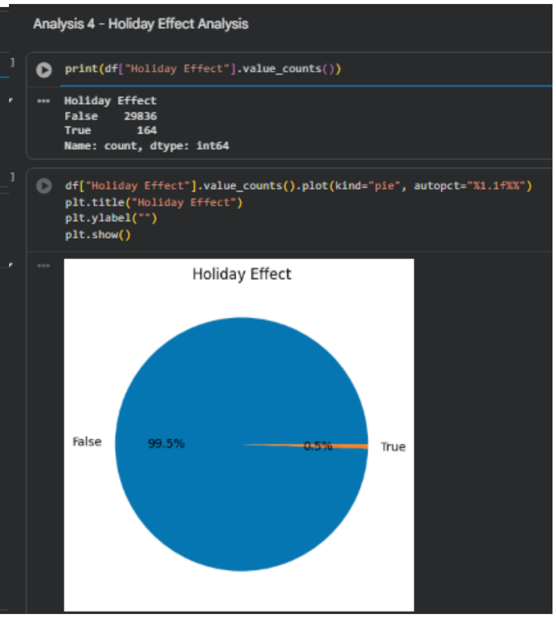
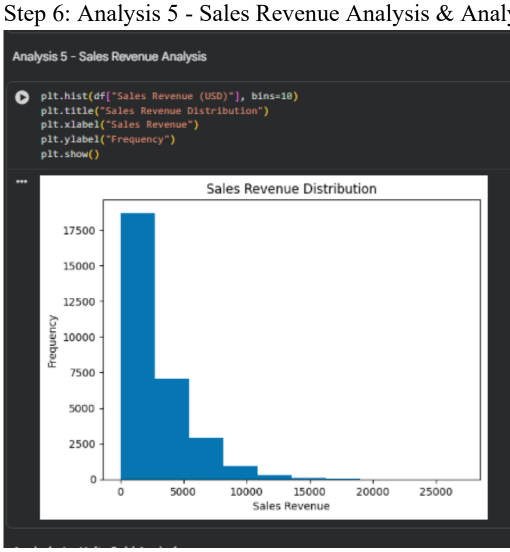
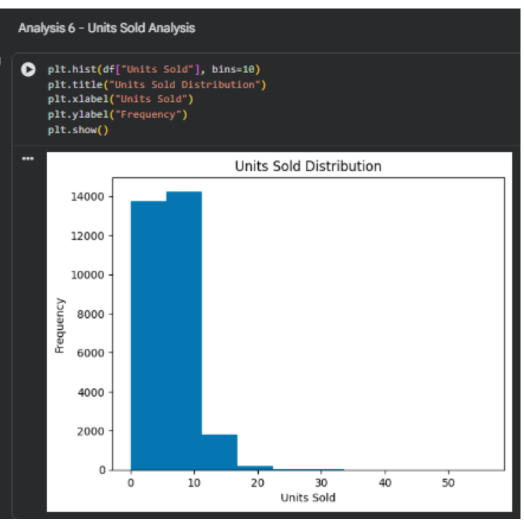




Step 6: Analysis 5 - Sales Revenue Analysis & Analysis 6 - Units Sold Analysis





Step 7: Analysis 7 - Marketing Spend vs Sales Revenue & Analysis 8 - Correlation Analysis